I Ethics Statement

FineBench was developed with careful consideration of ethical aspects in dataset creation and evaluation. Videos are used in accordance with appropriate licenses, and annotations should be used strictly for research purposes only. The benchmark is designed to evaluate fine-grained human action and interaction understanding, not to classify or infer sensitive personal attributes such as race, gender, or emotion. The dataset is released solely for academic research with the intention of advancing robust multimodal understanding.

II Reproducibility statement

To ensure the reproducibility of our work and spur adoption of our benchmark, we will release the dataset as a HuggingFace repository upon acceptance. We provide the code to reproduce our experiments using the VLMEvalKit [5] library on GitHub with explanations and a ready-made script to reproduce our experiments and run inference with any VLM supported in VLMEvalKit at this anonymous repository https://anonymous.4open.science/r/FineBench_eval-4AD4. We are also creating a leaderboard for FineBench and preparing a pull request to the official GitHub repository of VLMEvalkit after the review process is complete.

III Definition of "Fine-Grained" in FineBench

The term *fine-grained* is often used broadly in the vision-language literature to describe detailed or nuanced understanding. In this paper, we adopt a more specific meaning tailored to long-form video understanding. Concretely, *fine-grained* in FineBench refers to:

- **Spatial grounding**: questions are anchored to specific regions, objects, and persons within frames, enabling evaluation of models' ability to localize visual evidence rather than rely on global scene cues.
- **Temporal grounding**: questions reference precise temporal segments, sometimes spanning only a few frames, and require correct identification of when an action or interaction occurs within long videos (average length ~900s).
- **Human actions and interactions**: our QA pairs emphasize detailed human-centric phenomena such as person movements, person–person interactions, and person–object manipulations, going beyond coarse activity labels.
- **Compositional reasoning**: many questions require integrating multiple spatial or temporal cues (e.g., tracking an object across shots, or reasoning over sequential actions) to arrive at the correct answer.

Thus, "fine-grained" in FineBench does not imply exhaustive coverage of all possible aspects of granularity (e.g., attribute-level object categorization or subtle affective states), but specifically denotes the dense spatial and temporal grounding of human actions and interactions across extended video contexts. This operational definition aligns with the design choices of our dataset and clarifies the scope of our claims.

IV FAQs–Questions the authors have been asking each other

Why don't you evaluate very large open-source models (e.g., 72B)?. Running evaluations on such massive models requires substantial computational resources: GPUs, time, and ultimately money. Our goal was not to exhaustively benchmark every possible model size, but to identify meaningful trends across a representative spectrum of models. To balance comprehensiveness and feasibility, we included some of the strongest proprietary models available (e.g., GPT-4o, Gemini), which provide a useful reference point for what is achievable at scale. This strategy allows us to highlight the gap between cutting-edge proprietary systems and current open-source alternatives without incurring prohibitive costs.

How do you address possible data contamination? This is an important concern, especially since we leverage AVA (validation set) videos as a source. Unless one records new footage or secures exclusive copyright directly from producers, there is no way to guarantee that a given video has not been partially seen during pretraining. This is a limitation shared by virtually all modern benchmarks. However, our *annotations* are entirely original: we designed fine-grained, human-centric questions that did not exist before. More importantly, we are the first to explore dense, temporally precise VQA grounded in AVA at this level of detail. Thus, even if models were exposed to raw video data, the benchmark still tests capabilities that go far beyond memorization.

Why focus on fine-grained video understanding instead of broad, high-level comprehension? High-level scene understanding such as recognizing that "a person is cooking" is relatively mature in current VLMs. However, applications in assistive technologies, safety-critical systems, and human-computer interaction demand much finer distinctions: Did the person *deliberately* sit down, or did they *fall*? These nuanced judgments require compositional reasoning over subtle motion cues and temporal dynamics, which existing benchmarks largely ignore. Our benchmark is designed to push VLMs toward that more challenging and impactful frontier.

How is this benchmark different from existing video QA datasets? Prior benchmarks typically (i) rely on short clips, (ii) ask a small number of relatively coarse questions, or (iii) emphasize object-centric rather than human-centric reasoning. In contrast, FineBench provides nearly 200k QA pairs over long-form videos, with dense temporal coverage (over 3,100 questions per video). This density and human-centric focus uniquely enable rigorous stress-testing of VLMs' ability to reason about fine-grained human actions and interactions.

Can this dataset be used for training as well as evaluation? We design FineBench primarily as a benchmark, however the same strategy can be used to prepare the training data. In the future we will use a subset of that to create a dataset for few-shot finetuning.

What future directions does this benchmark enable? We see three main avenues: (i) developing specialized modules (like our FineAgent) to boost fine-grained reasoning in existing VLMs, (ii) improving temporal and spatial grounding techniques for multi-person settings, and (iii) fostering new datasets or synthetic approaches that complement our human-centric design. We hope FineBench serves as a catalyst for these next steps.

Why does Qwen2.5-VL benefit less from FineAgent? As shown in Table 4, Qwen2.5-VL (7B) exhibits only modest improvements when enhanced with FineAgent, especially compared to other open-source baselines. A natural question is whether this reflects a limitation of our framework or of Qwen2.5-VL itself. To investigate, we conduct an ablation study (Table 6) where we replace Qwen2.5-VL as the Descriptor with a different VLM (InternVL2.5-8B).

The results reveal that Qwen2.5-VL indeed benefits more when the Descriptor comes from a distinct model. Specifically, while the base Qwen2.5-VL improves by only +0.7 to +0.9 points when serving as its own Descriptor, substituting InternVL2.5 as the Descriptor yields larger gains of up to +1.4 points. This suggests that Qwen2.5-VL's weaker improvements are not due to FineAgent being ineffective, but rather due to limited complementarity when Qwen2.5-VL plays both roles simultaneously. In other words, FineAgent is most effective when the Localizer and Descriptor modules introduce genuinely new information beyond what the base model already encodes.

V Blind Evaluation of VLMs

To assess the contribution of visual input, we compare VLM's performance in its visual-aware mode against a blind version where only the textual (e.g., question) input is provided. As shown in Table 7, all models experience a substantial performance drop, highlighting the critical role of visual understanding in our benchmark.

VI Experimental Settings

Most of our experiments are conducted on 8 V100 GPUs (32GB each). Inference time vary widely across models with mPlug-Owl2 [30] and mPlug-Owl3 [29] taking around 2 hours, MiniCPM-V2.6 [28] taking 4 to 5 hours and InternVL2.5 [10] (7B) taking a little less than 15 hours per inference cycle. Trading off between enough context and computational complexity, we feed 5 frames to each model per question where the middle frame contains the main frame the question targets.

VII Data Creation Blueprint

In this section, we aim to give the reader a clear blueprint of how we create the dataset. The construction of our action recognition dataset employs a systematic methodology centered around the generation of Multiple Choice Questions (MCQs) derived from video action annotations. This blueprint outlines the core principles and stages of our data creation process, emphasizing the design choices made to ensure the dataset's quality and suitability for advancing research in action understanding.

Leveraging Structured Video Action Annotations. The foundation of our dataset lies in a collection of video sequences meticulously annotated with human actions. These annotations provide rich information, including temporal context, individual identities via bounding boxes, and corresponding action labels categorized into distinct types such as human movement, object interaction, and interpersonal engagement. This structured annotation framework serves as the primary source from which our question-answering pairs are generated.

Template-Guided Question Generation. To transform the action annotations into a queryable format, we utilize a carefully curated set of question templates, stratified by action category. As exemplified with comprehensive listings in Figures 4b, 4a, and 5a. These templates offer a standardized yet flexible approach to formulating questions about the observed actions. Furthermore, the spatial context of the actors within the video frame is considered to create more specific and grounded queries, enhancing the relevance of the generated questions.

Strategic Design of Distractor Options. A key contribution of our data creation process is the deliberate strategy employed for generating incorrect answer options (distractors). Recognizing the importance of challenging yet plausible alternatives, our methodology prioritizes semantic relatedness. We utilize a knowledge resource of similar actions (illustrated in Table 9) to generate distractors that closely resemble the correct action, thereby demanding a finer-grained understanding of the visual content. When semantically similar options are limited, the system draws from other actions within the same category and, as a final measure, from the broader action vocabulary, ensuring a comprehensive set of choices for each question.

Addressing Complex Action Scenarios. Our blueprint also accounts for instances where individuals perform multiple concurrent or related actions within a single video frame. In such cases, our methodology aggregates these actions into a single, more complex question. The corresponding correct answer reflects the combination of these actions. The distractors for these multi-action questions are designed to maintain a similar level of complexity, often involving combinations of other actions or partial overlaps with the correct set, thus promoting a deeper level of reasoning about the observed activities.

Ensuring Scalability and Reliability. The entire data generation pipeline is engineered for scalability, allowing for the efficient creation of a large-scale dataset from extensive video annotations. Furthermore, this rigid template approach contributes to the integrity of the dataset.

VIII The Subset

To facilitate more accessible experimentation for researchers with limited resources (ourselves included), we provide a highly representative smaller subset of FineBench. This subset is designed to enable rapid prototyping and low-cost benchmarking before scaling up to the full dataset, which is particularly beneficial given the high API costs of evaluating proprietary models. It consists of 20,143 QA pairs across 7 curated videos, amounting to approximately 10% of the full dataset.

Crucially, the 7 videos were deliberately selected to ensure the overall subset distribution closely mirrors the full foundational dataset across all critical metrics. As shown in Table 8, the subset strictly maintains proportional category distribution (with analogous ratios of Person Movement, Person Interaction, and Object Manipulation tasks) and question composition length. Furthermore,

Table 6: Ablation study using InternVL2.5 (8B) as Descriptor for FineBench inference with Qwen2.5-VL (7B). We report average accuracy (%) on FineBench. Each column corresponds to adding a specific module to the base VLM. Improvements over the base model are shown in green.

Model	+ Localizer		+ Descriptor		+ FineAgent	
	Acc.	Δ	Acc.	Δ	Acc.	Δ
Qwen2.5-VL (7B)	69.3	(+0.5)	69.5	(+0.7)	69.7	(+0.9)
Qwen2.5-VL (7B) †	69.3	(+0.5)	**69.9**	(+1.1)	**70.2**	(+1.4)

Object Manipulation

- What is {person} doing with the object?
- How is {person} interacting with objects in this frame?
- Which object-related action is {person} performing?
- How would you describe what {person} is doing with the item?
- What activity involving an object is {person} engaged in?
- How is {person} using the object?
- What function is {person} performing with this object?
- Which of the following describes {person}'s object interaction?
- What is {person} doing with their hands in this frame?
- Which task is {person} performing with the object?
- How is {person} using the object in this scene?
- What is {person}'s interaction with the item?
- Which action is {person} performing with the object?
- How would you describe the way {person} is handling the object?
- What is {person} doing to or with the item?
- Which of these describes how {person} is using the object?
- What object-based activity is {person} engaged in?
- How is {person} manipulating or using an item?
- Which type of object interaction is {person} demonstrating?
- What action involving an item is {person} performing?
- How would you characterize what {person} is doing with the object?

(a) Object Manipulation Templates

Person Movement

- What is {person} doing in this frame?
- How would you describe {person)'s movement?
- Which action is {person} performing?
- How is {person} positioning their body?
- What kind of body movement is {person} engaged in?
- How is {person} positioned in this frame?
- What physical activity is {person} performing?
- What posture or movement does {person} display?
- Which of the following best describes what {person} is doing?
- How would you characterize {person}'s physical position?
- What type of motion is {person} exhibiting?
- What physical state is {person} in?
- How would you characterize {person}'s posture?
- How is {person}'s body oriented in this scene?
- What is the stance or position of {person}?
- Which body posture is {person} showing?
- What kind of physical action is {person} engaged in?
- How would you describe the way {person} is moving?
- What bodily action is {person} performing?
- Which of these activities is {person} doing?
- What position has {person} taken in this frame?
- How would you categorize {person}'s physical activity?

(b) Person Movement Templates

Figure 4: QA templates for Object Manipulation and Person Movement

Person Interaction

- How is {person} interacting with another person?
- What type of social interaction is {person} engaged in?
- Which interpersonal action is {person} performing?
- How would you describe {person}'s interaction with others?
- How would you describe the interaction between {person} and the other individual(s)?
- What social behavior is {person} displaying?
- What social behavior is {person} exhibiting?
- How would you characterize the social dynamic between {person} and others?
- How is {person} acknowledging the other person?
- Which of the following describes how {person} is interacting?
- What kind of communication or contact is {person} having?
- How is {person} relating to others in this frame?
- What interpersonal activity is {person} involved in?
- Which social action is {person} performing?
- What is the nature of {person}'s interaction with others?
- How is {person} engaging with another individual?
- Which social gesture is {person} making?
- What type of interpersonal contact is {person} having?
- How would you characterize the way {person} is relating to others?
- What kind of social exchange is {person} participating in?
- How is {person} showing attention to the other individual (s)?
- Which of these describes the interaction {person} is involved in?
- What social behavior is {person} demonstrating?
- How is {person} communicating or connecting with others?

(a) Person Interaction Templates

Table 8: Key Statistics of FineBench Sub.

Statistic	Value
Total Questions	20,143
Unique Videos	7
Annotated frames/Video	806
Category Distribution:	
Person Movement	9645 (47.88%)
Person Interaction	6946 (34.48%)
Object Manipulation	3552 (17.63%)
Question Composition:	
Single Actions	16,709 (82.95%)
Combined Actions	3434 (17.05%)

as clearly evidenced by our empirical results in Table 3, open-source models evaluated on the subset exhibit near-identical performance scores to those evaluated on the full dataset (for instance, MiniCPM scores 58.4% on the subset vs. 59.2% on the full dataset; InternVL-2.5

Table 7: Performance (%) of VLMs in visual-aware vs. blind (LLM-only) settings.

Model	Visual-aware	Blind
Qwen2.5VL (7B)	68.8	43.5
MiniCPM-v2.6 (8B)	59.2	29.9
InternVL-2.5 (8B)	67.1	33.0

and Qwen2.5-VL display identical matching). This strong correlation conclusively demonstrates that subset results stand as a reliable and directly comparable proxy for the full benchmark, validating its use as a standard evaluation tier for proprietary models.

IX Details of FineAgent

IX.1 The Localizer

The **Localizer** module in FineAgent is responsible for spatially grounding the entity referenced in a natural language question. Given a frame from the video and the question (e.g., "What is the person on the left doing?"), the Localizer identifies the region in the image corresponding to the referent (e.g., "the person on the left") and produces a segmentation mask and pseudo-bounding box. This localized region is then passed to the VLM at inference.

To achieve this, we leverage the vision-language segmentation model, EVF-SAM [32], which supports referring expression segmentation using foundation models. The input question is first parsed using rule-based patterns to extract referential phrases. These phrases guide the segmentation model, which combines features from a BEiT-3 [22] language-image encoder and a Segment Anything Model (SAM) [11] backbone.

The Localizer outputs a binary segmentation mask and its corresponding bounding box, both of which are used to crop or mask the input image for further reasoning. If no valid region is found (e.g., due to ambiguous reference or occlusion), the system falls back to global reasoning using the entire frame. This spatial grounding module allows FineAgent to dynamically adjust its perceptual field based on linguistic cues, improving fine-grained video understanding accuracy.

IX.2 The Descriptor

The **Descriptor** module in FineAgent generates semantically meaningful captions from raw visual inputs, serving as a bridge between perceptual data and language-based reasoning. Given an image frame, the module produces a concise textual summary that captures the core scene content.

We use a Qwen2.5-VL (7B) accessed via the VLLM framework. Each frame is processed in isolation, where the model is prompted with the general instruction: *"Briefly describe this image."* This prompt is applied uniformly across all inputs to produce captions for each frame.

We intentionally omit prompts that elicit detailed scene annotations through highly specific prompts (e.g., descriptions of body posture or social interaction). This design choice is driven by a desire for *generalizability*: fine-grained prompts often induce brittle behaviors that are tightly coupled to training distributions, whereas general prompts yield more transferable representations and are more model-strength agnostic.

IX.3 Model Evaluation on FineBench

To evaluate the performance of VLMs, we constructed a dedicated evaluation pipeline tailored to the characteristics of our proposed dataset, **FineBench** into the VLMEvalkit [5] framework. Each example in FineBench consists of a stream of temporally ordered frames, optionally accompanied by spatial cues (bounding boxes in the case of FineAgent), and a multiple-choice question requiring nuanced temporal reasoning.

Streaming vs. Windowed Mode. We support two data presentation paradigms. In *streaming mode*, models are presented with a causal sequence of frames leading up to a target moment, simulating a real-time inference scenario. In contrast, the *windowed mode* provides a symmetric temporal window centered on the query frame, useful for evaluating models under less temporally constrained settings. In early experiments, we found that the windowed mode slightly outperforms the streaming mode (55.6 vs 55.5 for mPlug-Owl3), therefore, we stick with the former in this paper. However, both pipelines are supported and are open-sourced to allow other researchers to experiment with them. They also have similar performance and can be interchanged.

Caption and Spatial Grounding. When captions are available and enabled, they are extracted using timestamp alignment and included as additional input. Furthermore, if bounding boxes are annotated, models are prompted to attend specifically to the corresponding visual region in the target frame, supporting evaluations on spatial reasoning.

Prompt Construction. To probe model understanding effectively, we construct prompts that emulate a multimodal dialogue. Each prompt comprises a sequence of visual and textual components. Visual context is provided via a series of temporally ordered image frames extracted around the question timestamp. These are delivered either as a sliding buffer of preceding frames (streaming mode), or a centered clip with symmetric context (window mode). The number of frames (typically 8) and their stride are configurable.

Textual prompts follow a structured template designed to focus model attention on the correct frame. For example, in streaming mode (default), the prompt reads:

> These are the frames of a video stream. Select the best answer to the following multiple-choice question based on the most recent frame (frame t of N). Respond with only the letter (A, B, C, or D) of the correct option.

In the presence of captions, an additional block of dialogue text is inserted before the question prompt, reflecting the temporal alignment between frame timestamps and captions.

Furthermore for spatial grounding, we optionally append bounding box coordinates corresponding to relevant regions in the final frame:

> Pay attention to the region within the bounding box coordinates [x_min, y_min, x_max, y_max] in the target frame.

The final part of each prompt contains a clearly formatted multiple-choice question, e.g.:

Table 9: Examples of Similar Action Categories

Action	Similar Actions
bend/bow (at the waist)	walk, crouch/kneel, sit
dance	walk, run/jog, jump/leap
fall down	lie/sleep, get up, crouch/kneel
lie/sleep	sit, fall down, crouch/kneel
run/jog	walk, jump/leap, stand
carry/hold (an object)	lift/pick up, put down, throw
catch (an object)	throw, carry/hold (an object), lift/pick up
chop	cut, cook, stir
climb (e.g., a mountain)	walk, run/jog, jump/leap
close (e.g., a door, a box)	open (e.g., a window, a car door), pull (an object), push (an object)
eat	drink, cook, smoke
enter	exit, open (e.g., a window, a car door), close (e.g., a door, a box)
hit (an object)	kick (an object), throw, touch (an object)
lift/pick up	carry/hold (an object), put down, pull (an object)
open (e.g., a window, a car door)	close (e.g., a door, a box), pull (an object), push (an object)
paint	write, draw, touch (an object)
play board game	play with kids, play with pets, touch (an object)
press	push (an object), touch (an object), turn (e.g., a screwdriver)
pull (an object)	push (an object), lift/pick up, carry/hold (an object)
put down	lift/pick up, carry/hold (an object), throw
shovel	dig, lift/pick up, pull (an object)
text on/look at a cellphone	work on a computer, answer phone, read
throw	catch (an object), lift/pick up, carry/hold (an object)
touch (an object)	point to (an object), carry/hold (an object), press
fight/hit (a person)	martial art, kick (a person), push (another person)
give/serve (an object) to (a person)	take (an object) from (a person), hand shake, carry/hold (an object)
grab (a person)	hug (a person), lift (a person), push (another person)
hand wave	hand clap, hand shake, point to (an object)
listen to (a person)	watch (a person), talk to (e.g., self, a person, a group), answer phone
push (another person)	fight/hit (a person), kick (a person), grab (a person)
sing to (e.g., self, a person, a group)	talk to (e.g., self, a person, a group), listen (e.g., to music), play musical instrument
take (an object) from (a person)	give/serve (an object) to (a person), carry/hold (an object), grab (a person)
talk to (e.g., self, a person, a group)	listen to (a person), watch (a person), sing to (e.g., self, a person, a group)
watch (a person)	listen to (a person), talk to (e.g., self, a person, a group), take a photo

Question: What is the man doing in the last frame?
A. Walking out the door
B. Sitting on the couch
C. Pouring a drink
D. Looking at the camera
Answer:

Evaluation Protocol. Same as other Video MCQs in VLMEvalkit (using exact matching or rule-based to extract the correct letter).

Overall, this setup ensures that models are assessed not only for answer accuracy, but also for their ability to integrate multimodal cues under temporal constraints, which is the core challenge posed by FineBench.

IX.4 Qualitative Results and Analysis

To provide a more nuanced understanding of model performance, common challenges, and the impact of FineAgent, we present qualitative examples from FineBench in Figure 6 and Figure 7.

Figure 6 showcases scenarios where the base VLM (mPlug-Owl3) initially makes errors in identifying subtle human postures, positions, or the nature of object interactions. With the assistance of FineAgent, which provides enhanced spatial localization and descriptive context, the model is often able to correct its initial misjudgment. For instance, FineAgent helps distinguish between “Bend/bow" and “Stand," aids in differentiating “Sit" from “Crouch/kneel," and assists in understanding the continuous nature of an action (“Carry/hold") versus its initiation (“Lift/pick up"). These examples, consolidated in the figure, underscore the value of the Localizer and Descriptor components in FineAgent for tackling fine-grained tasks.

Figure 7 presents a comparison between two open-source models, Qwen2.5-VL (7B) and InternVL-2.5 (7B). The figure highlights a common failure mode: when faced with a question requiring precise identification and analysis of a specific individual in a crowded scene (e.g., “the 5th person from the left"), both models struggle,

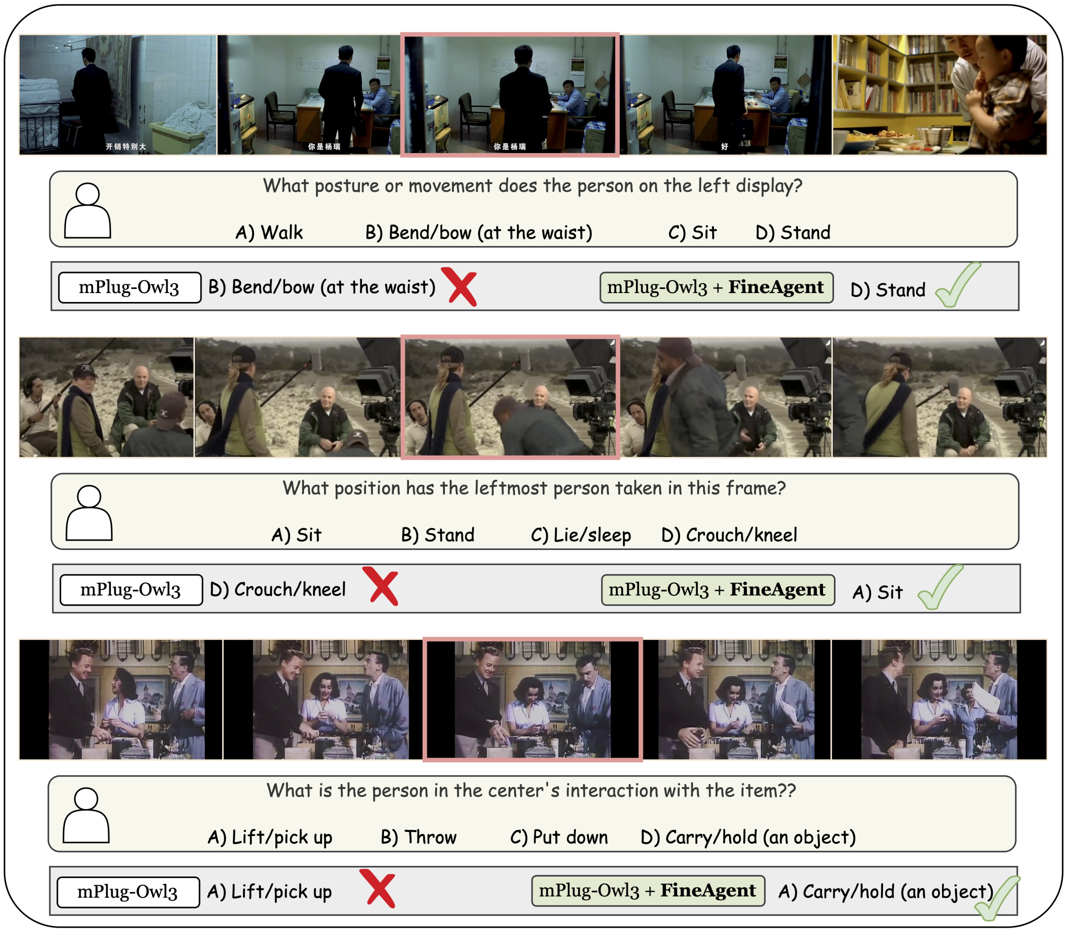

Figure 6: Qualitative examples demonstrating the improvements provided by FineAgent when applied to the mPlug-Owl3 model. The image highlights instances where the base model struggles with fine-grained distinctions in posture (e.g., distinguishing "Stand" from "Bend/bow"), position (e.g., "Sit" from "Crouch/kneel"), or nuanced object interactions (e.g., "Carry/hold" vs. "Lift/pick up"). With FineAgent's enhanced spatial grounding and contextual descriptions, the model often corrects these initial misjudgments, leading to accurate predictions.

selecting incorrect actions. This points to the persistent challenges in spatial reasoning and subject disambiguation that FineBench is designed to test, and which FineAgent aims to alleviate.

IX.5 Ethical Considerations and Broader Impact

Human-centric activity recognition inherently possesses a dual-use nature, necessitating a careful examination of its ethical implications. On the beneficial side, the capacity for machines to achieve a nuanced, fine-grained understanding of human actions enables transformative positive applications. As emphasized throughout this work, our explicit focus is on facilitating ambient intelligence systems that improve human well-being. These include advanced assistive technologies that offer greater independence for individuals with disabilities, reliable assisted living monitoring that accurately distinguishes between a deliberate action and a fall, and safe, intuitive human-robot collaboration in both industrial and domestic settings.

However, the analytical power required to interpret detailed human behavior also raises vital ethical challenges. We explicitly reject the use of such fine-grained recognition models for unwarranted monitoring or any applications that infringe upon human privacy. To mitigate the risks associated with this dual-use technology, we strongly advocate for the implementation of rigorous privacy safeguards. Deployed systems should utilize edge computing to process data locally without storing raw video, employ identity-anonymization techniques such as irreversible blurring or skeleton-only extraction, and strictly operate under continuous, informed consent from all subjects involved. Furthermore, as models become more adept at interpreting nuanced human interactions, the risk of perpetuating or amplifying social biases present in the training data becomes more pronounced. Robust guidelines, regular audits for bias, and transparent reporting are crucial. We believe

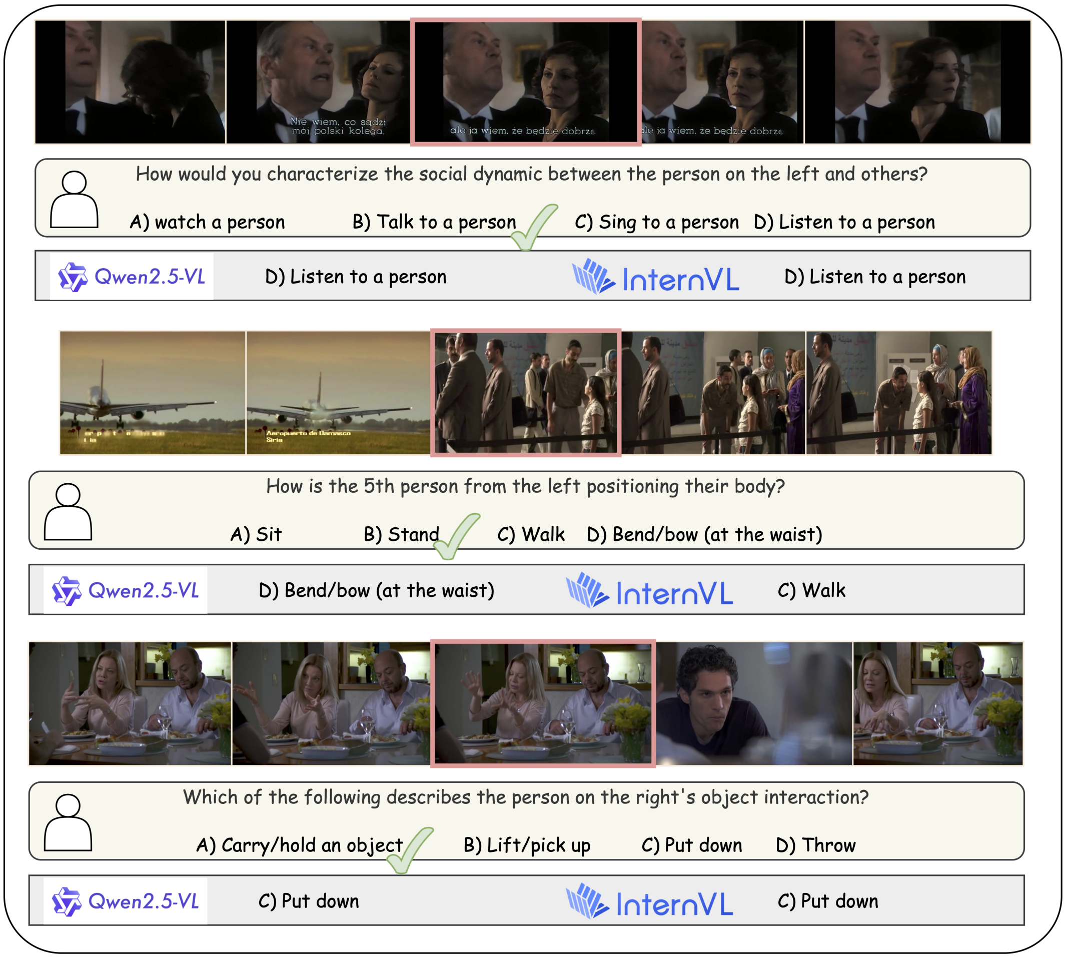

Figure 7: Qualitative examples comparing Qwen2.5-VL and InternVL on FineBench. The image illustrates scenarios including: (Top) incorrect identification of social dynamics ("Listen to a person") by both models; (Center) a common failure case where both models struggle with complex spatial referencing and posture identification in a crowded scene (e.g., "5th person from the left"), resulting in incorrect action predictions ("Bend/bow" or "Walk" instead of likely "Stand"); and (Bottom) an instance of incorrect object interaction identification ("Put down") by both models. These examples showcase the persistent challenges for VLMs in fine-grained understanding.

that fostering research in fine-grained understanding is vital, provided it remains strictly coupled with ethical deployment practices that prioritize human dignity and privacy.

IX.6 Limitations

While FineBench provides a valuable resource for assessing fine-grained human-centric video understanding and FineAgent offers a promising direction for enhancing VLMs, we acknowledge several limitations that also point towards avenues for future research.

Limitations of FineBench. The FineBench benchmark, despite its scale and density, has limitations primarily stemming from its reliance on the AVA v2.2 dataset, which may introduce biases from cinematographic content and define the scope of "fine-grained" actions. The use of predefined templates for question and distractor generation might not fully capture natural linguistic diversity and could lead to models exploiting patterns, while the multiple-choice format primarily tests discriminative rather than generative understanding. Although featuring long videos, the 64 unique video sources might also limit broad generalization across diverse real-world scenarios compared to datasets with a wider array of distinct video origins.

Limitations of FineAgent. The proposed FineAgent framework, while effective, presents certain limitations inherent to its modular design. These include the potential for error propagation from its Localizer or Descriptor components to the main VLM, and an increase in computational overhead and latency due to the sequential processing of these modules. The overall performance of FineAgent is also dependent on the capabilities of the underlying foundation models used for localization and captioning, and its effectiveness can be sensitive to prompt engineering and the strategy chosen for integrating auxiliary information. Lastly, FineAgent is specifically tailored to improve spatial reasoning and nuanced human action interpretation, and may not address other potential weaknesses in VLMs, such as very long-range temporal reasoning or abstract conceptual understanding.

Future work could focus on expanding FineBench with more diverse video sources and human-authored questions to mitigate template effects.

IX.7 Licence

AVA [9] is licensed under CC BY 4.0, and our annotations are under the MIT Licence.